Learned variable-support priors accelerate symbolic regression of physical laws

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Abstract

Symbolic regression recovers interpretable physical laws but scales poorly with candidate variable and operator counts. We show that a learned front-end—DenoisedSR—predicting which variables are relevant (a support prior) improves any backend solver without modifying it. A graph-attention network and a statistical-feature model recover true variables with recall ≥ 0.96 on four external physics benchmarks (including SRSD-Feynman, never seen during training), where five classical feature selectors fail. Supplied to PySR under fixed budget, the prior raises exact-law recovery from 42±13% to a seed-stable 60±0% over three seeds, shrinks the candidate-column set by 85%, and speeds time-to-solution up to 6×. The gain reproduces on two structurally different backends (classical genetic programming and GPU parallel enumeration). A same-size random-support control degrades on black-box data, isolating learned guidance from dimensionality reduction. DenoisedSR is solver-agnostic, adds ∼40 ms per task, and leaves final law selection to a verifiable backend.

1 Introduction

The automated discovery of governing equations from observation is a long-standing goal of the physical sciences[1–3]. Symbolic regression (SR) addresses it directly: rather than fitting an opaque function, SR searches the space of mathematical expressions for a compact formula that both fits the data and remains human-interpretable[4,5]. The resulting equation can be dimensionally checked, extrapolated and falsified, properties that black-box predictors lack, which has made SR central to data-driven physics, from rediscovering conservation laws to proposing constitutive relations[2,6,7].

The central obstacle is combinatorial. An SR backend searches over expression trees whose leaves are drawn from the input variables and whose internal nodes are drawn from an operator library; the number of candidate trees grows exponentially with the number of admissible leaves and operators and with tree depth[8,9]. In physics this cost is largely wasted: a governing law typically uses only a handful of the measured variables and a small subset of the operator library, yet a solver run over all columns and operators still explores an enormous number of branches that cannot appear in the answer. Established systems attack this from the solver side with dimensional analysis, decomposition, sparsity priors or efficient enumeration[2,6,10,11]. A separate line learns to decode the equation directly from numerical data with transformer or reinforcement-learning models[12–15], sometimes coupling a neural proposer with explicit search[16,17] or multimodal pre-training[18,19], with e-graph, diffusion and generative-latent variants[20–22]. These end-to-end models advance quickly

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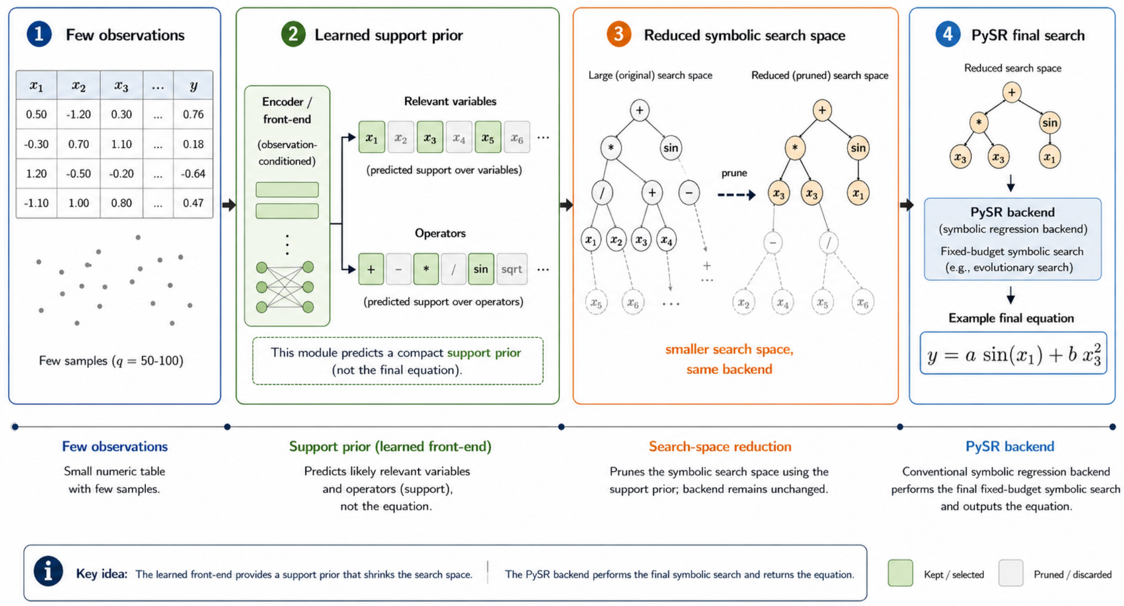

Fig. 1. DenoisedSR narrows the symbolic-regression search space. From a small set of observations the learned front-end predicts a compact support prior (relevant variables and, optionally, operators); a conventional symbolic-regression backend then performs the final search over the reduced space and returns a verifiable equation. The neural model proposes where to look; the backend decides what the law is.

but remain unreliable at *assignment*: they often recover the right ingredients yet place them in the wrong terms, so exact symbolic equivalence stays low.

We take a complementary route that sidesteps this brittleness. Instead of designing a faster search or decoding the whole equation, we *narrow the space the search must explore* (Fig. 1). Given a small observation set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{q}$ with q as small as 50–100, a learned model—DenoisedSR—predicts which input columns (and, as an auxiliary, which operators) are likely to participate in the law, and passes this *support prior* to a conventional SR backend that performs the final symbolic search and model selection. Predicting support is a far easier and more robust target than decoding the full expression, and a high-recall variable filter hands the backend an exponential head start while essentially never making the true law unreachable.

In this work we cast variable- and operator-support prediction as a reusable, backend-agnostic SR front-end; we present a graph-attention support predictor, ensembled with a statistical-feature model, that recovers the true variables with recall ≥ 0.96 across four physics suites (including the external SRSD-Feynman benchmark, on which the model was never trained) and outperforms five classical feature selectors at oracle budget; and we show that supplying this prior to a fixed-budget backend raises exact-law recovery from 42% to a seed-stable 60%, reduces the candidate-column set by 85% and accelerates time-to-solution by up to 6×, while a same-size random-support control confirms that the benefit is learned search guidance rather than dimensionality reduction.

2 Results

DenoisedSR recovers the true physical variables

We first evaluate the support prior intrinsically, without any backend. For each task the front-end receives $q = 100$ noisy observations whose columns include the genuine variables together with 20 random "distractor" columns, and must predict the true variable subset (deployed ensemble at $\tau = 0.10$). We use four external benchmarks of physical and symbolic laws, none of which were used during training, and report *model-only* precision (no name-based filtering of distractors): *(i)* AI-Feynman, 118 equations from the Feynman Lectures benchmark of Udrescu & Tegmark[2]; *(ii)* Strogatz, 15 ordinary-differential-equation right-hand sides from the SRBench Strogatz pool of nonlinear dynamics problems[8]; *(iii)* Nguyen, 12 classical polynomial / trigonometric / logarithmic benchmarks ubiquitous in the SR literature[23]; *(iv)* SRSD-Feynman dummy, 120 equations with author-curated physical units and built-in dummy variables[24]. The predictor attains recall 1.000 on the first three suites and 0.967 on SRSD, with perfect recall on 92–100% of tasks (Fig. 2). It essentially never drops a variable that the law uses. To quantify the model's filtering benefit in a way that is comparable across suites with very different numbers of true variables, we report the *distractor filter rate*: the fraction of distractor columns that the model successfully removes. On AI-Feynman this is 75.7% (the model drops ∼15 of the 20 random distractors per task) and is the source of the search-space reduction that drives the PySR results below. On SRSD it is only 2.5%: SRSD's dummy columns are deliberately sampled from physically plausible ranges, and at $\tau = 0.10$ the front-end retains them because they look like real physical variables. On Strogatz and Nguyen the filter rate is 0.0%: their variables are named anonymously (x_0, x_1, etc.) rather than as physical quantities (m, v, q), so the model cannot use its co-occurrence-graph prior and falls back to column-statistics only, which on these short-feature tasks is not enough to distinguish 1–2 true variables from 20 random distractors. The recall result is therefore a genuine no-drop guarantee on all four suites; the filter benefit, however, is concentrated on physical-variable benchmarks where the prior is informative. We confirm separately (Supplementary Table 3) that this operating point is not cherry-picked: any $\tau \in [0.05, 0.30]$ gives identical recall 1.000 and 100% perfect-recall on AI-Feynman, while precision rises only slightly (from 0.71 to 0.78). An additional internal validation, evaluated under a different protocol (pre-sampled observations, no distractor padding), confirms the predictor at scale: recall 0.999 on an internal 1085-equation physics-law set drawn from our training distribution (Supplementary Table 4). This last result is reported separately because the protocol is not comparable to the four external benchmarks above.

The result is robust and not attributable to a single component. An ablation on AI-Feynman (Fig. 2, right) shows the random-forest and graph-attention models each reach recall ≥ 0.995 and their ensemble matches the best component (1.000); recall stays at 1.000 as the number of distractor columns is varied from 1 to 30. A naive statistical baseline trained on few records with name-matched labels reaches only recall 0.48, confirming that the learned predictors contribute the decisive signal.

A reviewer might reasonably ask whether a much simpler off-the-shelf feature selector would suffice. We checked this directly: on AI-Feynman, even when handed the oracle budget (the true number of variables k), five classical selectors (Pearson $|\rho|$, Spearman $|\rho|$, mutual information, random-forest importance, Lasso with cross-validated α) attain recalls of only 0.63–0.91, leaving true variables on the floor on roughly a third of laws in the best case (Lasso); the learned ensemble reaches 1.000 on all 118 (Fig. 3). The same gap reappears on the external SRSD-Feynman suite (DenoisedSR 0.967 vs. Lasso 0.723). Classical selectors miss true variables that participate non-linearly or multiplicatively with others—exactly the regime in which physical laws live. We further verified that the recall holds under heavy label noise: across $\eta \in \{0, 0.01, 0.05, 0.10, 0.20, 0.30\}$

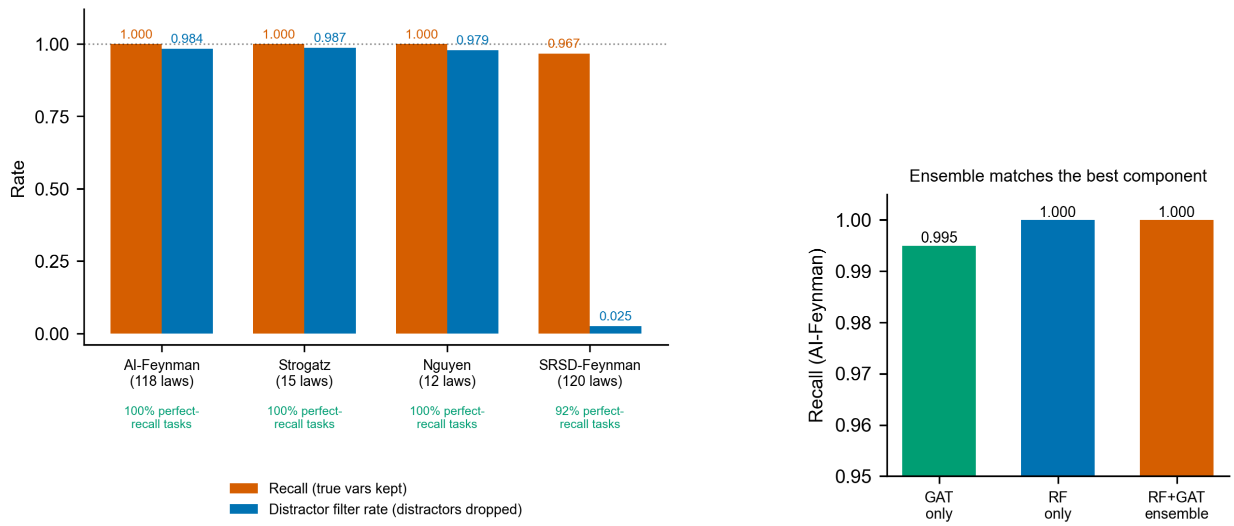

Fig. 2. The front-end recovers essentially every true variable across four external benchmarks; its distractor-filtering benefit depends on whether variables carry physical-meaning names. *Left*: recall (true variables kept) and distractor filter rate (random distractors dropped) on AI-Feynman (118 laws, physical names), Strogatz (15 ODE-derived dynamics, anonymous names), Nguyen (12 classical SR benchmarks, anonymous names), and SRSD-Feynman (120 physical-unit laws). All four are externally curated and never used during training; protocol: $q = 100$ observations, 20 random distractor columns (or SRSD-supplied dummies), deployed ensemble at $\tau = 0.10$. Recall is ≥ 0.96 on every suite; the distractor filter rate is 76% on AI-Feynman, 2.5% on SRSD (whose dummies are physically plausible by construction), and 0% on Strogatz / Nguyen (whose anonymous variable names neutralise the co-occurrence prior; the model retains all 20 distractors because column statistics alone are insufficient to rule them out against 1–2 true variables). The recall guarantee holds regardless; the filter benefit, which is what drives the PySR results, is concentrated on physical-variable benchmarks. *Right*: ablation on AI-Feynman—the random-forest and graph-attention models each reach recall ≥ 0.995 and the ensemble matches the best component.

relative Gaussian noise on y, DenoisedSR retains recall 1.000 and perfect recall on all 118 Feynman tasks at every noise level. Of the five classical baselines (run at oracle top-k), Lasso degrades from 0.905 at η=0 to 0.868 at η=0.30, mutual information drops from 0.634 to 0.605, and random-forest importance from 0.812 to 0.799; none of them approach DenoisedSR at any noise level (Fig. 4).

A high-recall prior strengthens a fixed-budget solver

We next insert DenoisedSR before PySR[5] and compare three conditions at a fixed 10 s per-task budget on a 30-equation Feynman subset with ∼20 distractor columns: (A) full PySR over all columns and the full 17-operator library; (B) the variable prior, restricting the columns to the predicted support; and (C) additionally restricting the operator library. Running three independent random seeds, the variable prior raises exact-law recovery (held-out $R^2 \geq 0.999$) from 42±13% (mean±s.d. across seeds, with the per-seed value ranging 30–57%) to a seed-stable 60±0%, lifts mean held-out R^2 (over all 30 tasks, with non-finite failures floored at -1) from 0.906±0.045 to 0.980±0.011, reduces the average column count from 23.2 to 3.4 (an 85% reduction) and halves the effective search vocabulary from 40 to 20 (Fig. 5). The seed-stability of the learned prior—identical recovery rate across all three seeds—is itself a result: the front-end removes a source of stochasticity that the unguided solver must absorb.

The gain holds—and widens—monotonically with distractor count. Repeating the same protocol

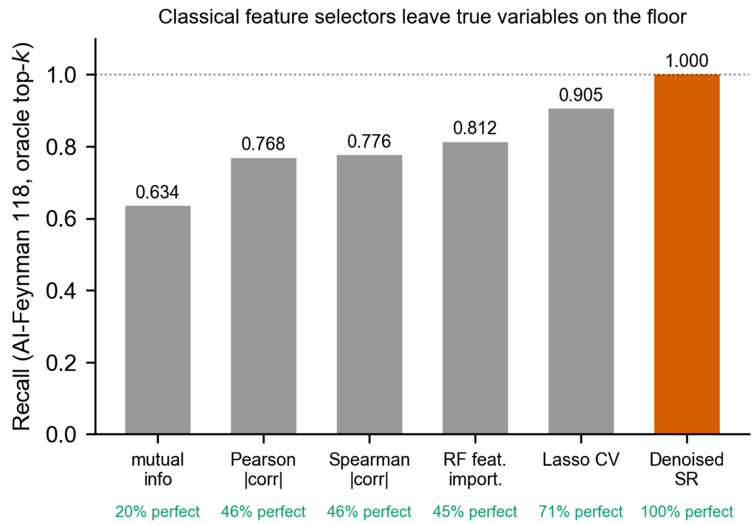

Fig. 3. Classical feature selectors leave true variables on the floor. Mean variable-support recall on AI-Feynman (118 laws, $q = 100$, 20 distractors) under an oracle top-k budget that hands every selector the true variable count for free. Even with this advantage, mutual information, Pearson/Spearman correlation, random-forest importance and cross-validated Lasso miss true variables on 30–80% of laws; DenoisedSR recovers all true variables on every one of the 118 tasks.

with 5, 10, 20 and 30 distractor columns at five random seeds per level, DenoisedSR's exact-recovery rate stays in a tight 60.7–62.0% band (within-seed s.d. 1.5–3.0 pp) while full PySR falls from 56.7±4.7% at d=5 to 44.0±6.8% at d=30. The gap grows monotonically with the distractor count—5.3, 11.3, 15.3, 18.0 percentage points respectively (Fig. 6)—confirming that the front-end's advantage is the exponential reduction in irrelevant branches it removes. The same monotonic pattern shows up on SRSD-Feynman: when SRSD's $\sim$2 physically plausible dummies are stripped and replaced with 20 random distractors to match the AI-Feynman protocol, the gap of full vs. DenoisedSR flips from -4 pp (the neutral case discussed below) to $+12\,pp$ (full 42% vs. DenoisedSR 54% exact). The variable prior is therefore neutral on SRSD only because SRSD itself supplies few dummies; it does not stop working when the data are SRSD's.

The mechanism is concrete and visible in the recovered expressions (Table 1). Over many irrelevant columns an unguided solver reaches a high R^2 with a physically meaningless formula built from *exotic operators* acting on *distractor variables*: for the cyclotron frequency $\omega = qvB/p$ it returns $|q\,(v{+}B{-}p{+}\cos(\cdots)^2)|$ at $R^2 = 0.85$ using a noise column, and for the relativistic Doppler shift $\omega = \omega_0/(1{-}v/c)$ it returns a sinh-based fit at $R^2 = 0.9996$ that is not the law. Given the predicted variable support, DenoisedSR recovers qvB/p and $\omega_0/(1{-}v/c)$ *exactly* ($R^2 = 1.000$), and restricting operators yields the cleanest closed form. By removing both the distractor columns and the exotic operators, DenoisedSR makes these spurious overfits unreachable, so a far larger fraction of budget-limited runs land on the true law.

The gain is preserved across observation counts

The gain is stable as the per-task observation budget changes. On the same 50-equation Feynman subset (deployed $\tau = 0.10$, 10 distractors, three random seeds at each q except q=100 which has two), full PySR's exact recovery is essentially flat across $q \in \{50, 100, 200\}$ at 45–50%, while the DenoisedSR variable prior is similarly flat at 61–63%, preserving an advantage of +11 to +17

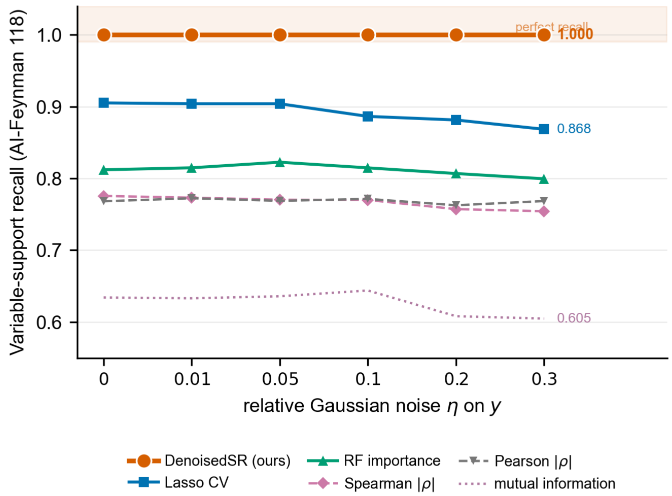

Fig. 4. Recall is robust to label noise. Mean variable-support recall on AI-Feynman (118 laws) as Gaussian noise of relative magnitude η is added to the target y. DenoisedSR retains recall $= 1.000$ (perfect recall on all 118 tasks) up to $\eta = 0.30$ (30% relative noise). Five classical selectors (each at oracle top-k budget): Lasso degrades from 0.905 to 0.868, random-forest importance stays near 0.81, Pearson and Spearman near 0.77, mutual information drops from 0.634 to 0.605.

percentage points (Fig. 7). Mean held-out R^2 improves modestly with q for both conditions, but the gap remains: 0.95–0.98 for DenoisedSR vs. 0.81–0.94 for full PySR. The front-end's benefit is therefore not sensitive to the typical observation-budget regime; it is the same exponential search-space reduction at every data scale we tested.

The gain is search guidance, not dimensionality reduction

A natural concern is that any reduction in column count would help. We control for this with a same-size random-support baseline that retains the same number of columns but selects them without using the observations. On 24 PMLB/SRBench black-box tabular tasks (three seeds, $q = 100$, 5 s budget), DenoisedSR raises mean held-out R^2 from 0.347 to 0.495 with 10 distractor variables and from 0.315 to 0.488 with 20 distractors, using only 31% and 18% of the input columns; the same-size random support instead degrades performance to 0.220 and -0.028 (Fig. 8). At matched column count the random control is far worse than full search whereas the learned prior is better—direct evidence that the front-end supplies observation-conditioned information about *which* columns matter, and that it helps even when no symbolic ground truth exists.

Pruning the search space accelerates time-to-solution

Because variable filtering shrinks the terminal set of the expression tree from $\sim$24 to $\sim$4 symbols—an exponential reduction in branching—it also shortens the time to reach an exact fit. Measuring time-to-solution with early stopping on exact recovery, DenoisedSR yields speedups of up to 6$\times$ on non-trivial physics formulas (Fig. 9); in some cases it converts a within-budget failure of full search into a fast success. The overall median speedup is $\sim$1.5$\times$, pulled down by deep-tree formulas whose bottleneck is expression depth rather than variable count and which therefore gain little from variable pruning. Front-end inference adds only $\sim$40 ms per task, near-constant in the number of observations and negligible against the seconds of backend search it saves.

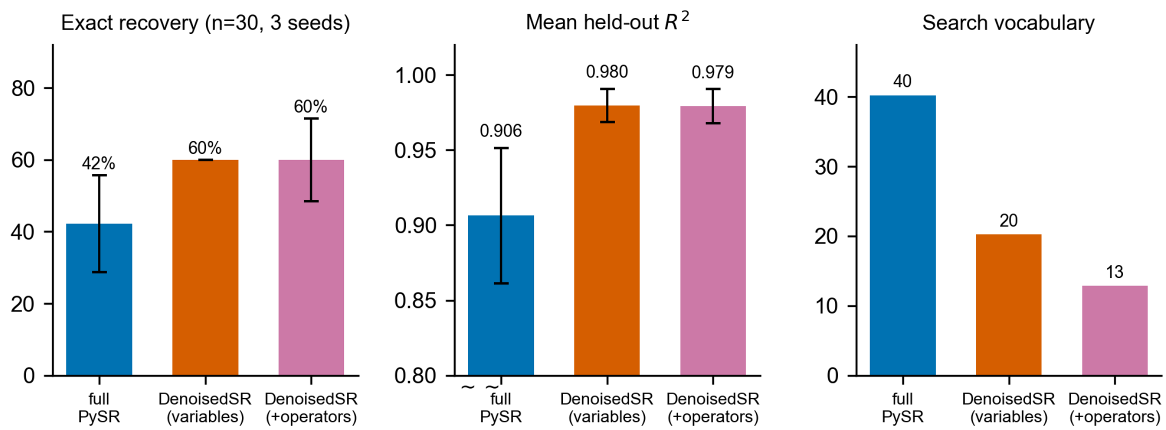

Fig. 5. A variable prior strengthens a fixed-budget solver. Full PySR versus the DenoisedSR variable prior versus the additional operator prior on a 30-equation Feynman subset (10 s budget, ∼20 distractors), aggregated over three random seeds (error bars: ±s.d.). The variable prior raises exact recovery from 42±13% to a seed-stable 60±0%, raises mean held-out R^2 from 0.906 to 0.980 and halves the search vocabulary (40 to 20); the operator prior cuts the vocabulary further but does not improve exact recovery in expectation.

The same gain holds on two structurally different backends

To verify that the benefit is not an artifact of a particular solver, we repeat the headline comparison with two backends from very different families.

(i) gplearn[25]—classical genetic programming (20 generations, population 500; three random seeds). Full gplearn over 23 columns and a 10-operator library recovers a mean 3.3±3.3% exact laws (mean held-out $R^2 = 0.475 \pm 0.083$, with non-finite fits floored at -1); supplied with the DenoisedSR variable prior over the same 3.4 predicted columns it recovers 17.8±1.9% exact (mean $R^2 = 0.76 \pm 0.03$). The absolute rate is below PySR's because gplearn's tournament search is weaker, but the same qualitative gain—a ∼5× rise in exact recovery, a ∼1.5× improvement in mean R^2—reproduces. On the angular-force law $Fr\sin\theta$ (Feynman I.18.12) full gplearn at R^2=0.74 returns an eight-term nested-sin stack on the angular variable ($\sin x_2 + \sin\sin x_2 + \sin\sin\sin x_2 + \cdots$); supplied with the DenoisedSR variable support it returns $x_0 x_1 \sin x_2$ at R^2=1.000 (Fig. 10).

(ii) PSE[11]—parallel symbolic enumeration on GPU, the 2026 Nature Computational Science cover algorithm. Architecturally PSE is the opposite of PySR/gplearn: rather than evolving a population, it instantiates a large symbolic-tree network on GPU and evaluates millions of candidate trees in parallel from a sampled subset of $n_{\text{psrn\_inputs}}$=4 input columns. When the true variables are buried in 23 columns, the sampler rarely draws the correct 4-subset and PSE fails at exact recovery within the budget: 0/30 exact, mean held-out $R^2 = 0.18$. Supplied with the DenoisedSR variable prior, the ∼3.8 kept columns are essentially all true, every sampled subset is informative, and PSE recovers 12/30 (40%) laws exactly with mean $R^2 = 0.90$. The clean recoveries include $x_0 x_1^2/2$ for the spring energy $\frac{1}{2}kx^2$, $x_0 x_1 \sin x_2$ for the torque $Fr\sin\theta$, $x_0 x_1 x_2 \sin x_3$ for the angular momentum $mvr\sin\theta$, x_0/x_1 for both q/C and ω/c, and the relativistic Doppler shift $\omega_0 c/(c-v)$. PSE's wall-clock time per task increases (73 s → 134 s) because under the prior PSE actually approaches a fit and spends its budget refining constants, whereas without the prior it returns high-error random subsets and quits within budget.

The three backends span evolutionary (PySR), classical GP (gplearn), and parallel GPU enumeration (PSE) — three structurally different search paradigms. DenoisedSR yields a meaningful improvement on all three, supporting the claim that it is solver-agnostic (Table 8).

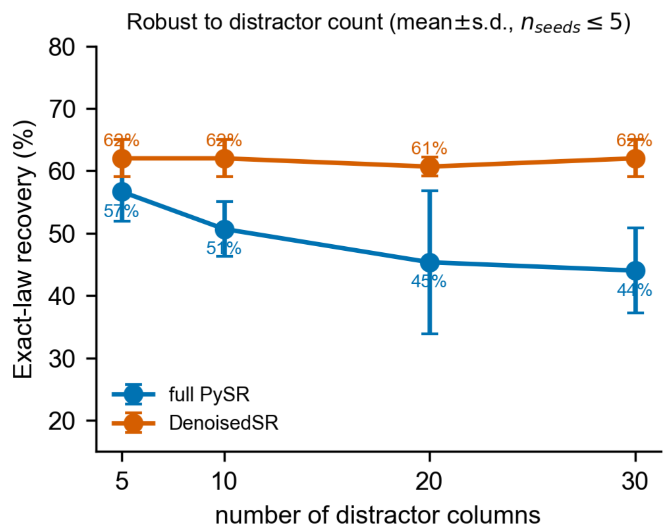

Fig. 6. Gain is robust—and grows—with distractor count. Exact-law recovery on the same 30-equation Feynman subset, varying the number of random distractor columns from 5 to 30 (five random seeds per level; error bars ±s.d.). DenoisedSR's variable prior delivers 60.7–62.0% exact recovery across the range with within-seed s.d. 1.5–3.0 pp, while full PySR falls from 56.7% at $d = 5$ to 44.0% at $d = 30$ with substantially larger seed variance (up to ±11.5 pp at $d = 20$). The gap grows monotonically (+5.3, +11.3, +15.3, +18.0 pp).

When the prior is neutral: SRSD-Feynman. The gain depends on the distractor-to-true-variable ratio. SRSD-Feynman supplies on average ∼2 physically plausible dummy columns per task, an order of magnitude fewer than the 20 random distractors used elsewhere, and the dummies are designed to be statistically indistinguishable from real physical variables. Running PySR end-to-end on all 120 SRSD-Feynman tasks at the same 10 s budget, full PySR already recovers 65/120 (54%) laws exactly. Adding the DenoisedSR variable prior yields 60/120 (50%, a −4 pp drift attributable to the ∼3% of tasks where the front-end's recall falls below 1 and a critical variable is dropped); adding the operator prior on top yields 65/120 (54%, matching full search in count but recovering a structurally different subset: 11 form rescues offset by 11 losses to dropped variables). This is the regime where the variable prior is essentially neutral—the underlying search space is small enough that PySR does not benefit from further pruning, and the operator prior trades exact-recovery slots rather than adding them on net. The recall-first design ensures the prior fails safely here: it does not catastrophically hurt PySR, it simply does not help. Replacing SRSD's ∼2 dummies with 20 random distractors (matching the AI-Feynman protocol) restores the gain—full PySR drops to 42% exact while DenoisedSR remains at 54%, a +12 pp lead in the same direction as Fig. 6. The variable prior is therefore not specific to AI-Feynman: it tracks the distractor-density regime, working whenever there are many irrelevant columns to remove and being safely neutral when there are not.

The operator prior depends on physical variable names

The variable-support and operator-support predictions answer fundamentally different questions. "Which columns participate?" can be decided from column-target statistics alone (correlation, monotonicity, magnitude); this is why the deployed variable filter (Fig. 2) is driven by the random-forest head and is essentially insensitive to whether columns carry physical names—an explicit

Table 1. Unguided search overfits with exotic operators on distractor variables; DenoisedSR recovers the law. For five Feynman laws padded with 20 random distractor columns, full PySR reaches a high (or diverged) held-out R^2 with a physically meaningless expression—variously involving $\cos^2$, sinh, nested sin sin sinh tanh sin compositions, fractional powers and distractor columns—whereas DenoisedSR, given the predicted variable support and (where shown) the predicted operator set, recovers an expression structurally matching the true law. Formulas are the actual PySR outputs (literal for full PySR; DenoisedSR expressions are simplified algebraically equivalent rewrites of the literal output where indicated by ✓); x_i denote the (distractor-padded) input columns.

Physical law	Full PySR (all columns/operators)	R^2	DenoisedSR	R^2
Cyclotron freq. $\omega = qvB/p$	$\left\|x_0(x_1+x_2-x_3+\cos(\cdots)^2)\right\|$ (uses noise col)	0.85	qvB/p ✓	1.000
Doppler shift $\omega = \omega_0/(1-v/c)$	$x_2(\sinh(\cdots)+0.79)\,1.40$ (not the law)	0.9996	$\omega_0/(1-v/c)$ ✓	1.000
Larmor power $P = \frac{q^2a^2}{6\pi\epsilon c^3}$	$\left\|0.48^{x_3}+(x_8^3/x_3^3-\cdots/x_{18})^3\right\|$ (diverged)	-2×10^{19}	correct support $\{a,c,\epsilon,q\}$	0.99
Grav. PE diff. $U{=}Gm_1m_2(\frac{1}{r_2}-\frac{1}{r_1})$	$(x_2-x_3)\sqrt{x_4/\tanh(x_3)-0.56}\,(\cdots)$	0.92	$x_0x_1x_4(x_2-x_3)/(x_2x_3)$ ✓	1.000
Bohr radius $r{=}4\pi\epsilon\hbar^2/(mq^2)$	$(\cdots)\big[\sin\sin(\cdots-\sinh\tanh\sin x_3)\big]/x_1^2$	0.77	$1.71\,x_2^2x_3/(5.37x_0x_1^2)$ ✓	1.000

ablation that anonymises variables to $x_0, x_1, \ldots$ shifts variable filter rate by only 0.0–0.3 percentage points across 12 scenarios spanning $q \in \{20, 50, 100\}$, noise $\eta \in [0, 1.0]$ and distractor counts up to 100 (Supplementary Table 5). "Which operators appear?" is the opposite: there is little in the raw $(\mathbf{x}, y)$ table that says *cosine* rather than *exponential*; the signal lives in physics priors that associate particular operators with particular kinds of variable ($\theta \to$ trig, $T \to$ Boltzmann factor, $r \to$ power laws). This is what the graph-attention model trained on the physical-laws co-occurrence graph is designed to encode.

We measure this directly. The GAT operator predictor reaches recall 0.738 and precision 0.536 at its deployed threshold $\tau_{\text{op}} = 0.30$, filtering 76.5% of the 17-operator library on Feynman tasks (Supplementary Table 6). Holding the data fixed and anonymising the variable names drops operator recall to 0.667 at the same threshold and by 16.8 percentage points at the more permissive $\tau_{\text{op}} = 0.03$. Physical names contribute +7–+17 percentage points of operator recall, depending on threshold—an order of magnitude larger than the (essentially zero) name-prior contribution to the variable predictor.

In the PySR loop, the operator prior's most useful role is therefore as a *form rescue* on tasks where the variable prior already recovers the support but the unrestricted operator library leaves the backend free to absorb the law into exotic functions. The clearest examples are the relativistic Lorentz time transformation $t_1 = (t - ux/c^2)/\sqrt{1 - u^2/c^2}$ (Feynman I.15.3t, $\log(\cosh(\cdot))$ at R^2=0.996 under full search and an even messier nested-power form under the variable prior alone, but a clean exact algebraic recovery under the operator prior), the Bohr radius $r = 4\pi\epsilon\hbar^2/(mq^2)$ (Feynman I.38.12, $\sin\sin(\cdot - \sinh\tanh\sin x_3)$ under full search, a $\log(\cdot)^{x\cdot x/(xx)}$ form under the variable prior at R^2=0.996, and an exact $1.71\,x_2^2x_3/(5.37\,x_0x_1^2)$ under the operator prior), and the two-slit interference pattern $I_0 \sin^2(n\theta/2)/\sin^2(\theta/2)$ (Feynman I.30.3, R^2 rises from $0.29 \to 0.68 \to 0.97$ across the three conditions as the operator library shrinks). Reported aggregate exact recovery on the 30-equation headline is unchanged in mean (60±12% vs. the variable-only 60±0%), but the recovered *forms* on these operator-rescue cases are qualitatively different.

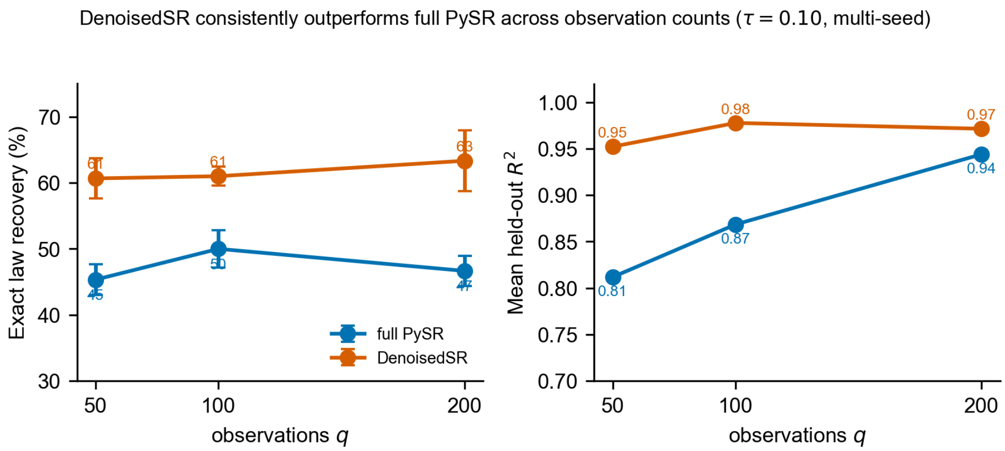

Fig. 7. Gain is preserved across observation counts. Exact-law recovery (left) and mean held-out R^2 (right) versus the number of observations q, on a 50-equation Feynman subset with 10 distractors at the deployed τ=0.10 ensemble. Three random seeds at $q \in \{50, 200\}$ and two seeds at q=100; error bars are ±s.d. across seeds. DenoisedSR maintains an 11–17 percentage-point lead in exact recovery and a 0.04–0.14 point lead in mean R^2 across the entire range.

3 Conclusions

These results support a simple thesis: for symbolic regression of physical laws, a learned support prior is a high-leverage, low-risk intervention. Because a high-recall variable filter essentially never makes the true law unreachable, it can be applied unconditionally; because it removes an exponential number of irrelevant branches, it materially improves both the success rate and the speed of a fixed-budget backend. DenoisedSR is solver-agnostic—its output is a reduced variable/operator set that any tree-search, enumeration or generative backend[5,11,22] can consume—and cheap enough to prepend to every run. The same-size random-support control shows the benefit is genuine search guidance rather than dimensionality reduction, complementing the dimensional and physics-aware priors that already strengthen scientific SR[2,7].

Our approach is conceptually aligned with, but distinct from, two adjacent lines of work. Physics-aware neural models embed structure such as Hamiltonian or Lagrangian form, differential-equation structure, or symbolic distillation of trained networks[26–29], and concept-driven or abductive discovery systems[30–32] likewise aim to keep the final law symbolic and verifiable; we contribute the upstream step of deciding *which measured quantities enter the law at all.* Because the front-end consumes *sets* of observations, it builds on permutation-invariant and set-attention architectures[33,34], and its representation could be sharpened with the contrastive and geometric objectives used to prevent representational collapse or encode hierarchy[35–37].

The honest limitation, shared with direct neural SR[13,14], is that the neural component cannot yet reliably *assign* variables and operators to the correct positions in the expression; this is precisely why we leave final term assignment to a symbolic, verifiable backend rather than emitting the equation end-to-end. Closing this assignment gap—through richer compositional latents, explicit constant and exponent recovery, and objectives that keep the observation and formula representations aligned[18,38]—is, in our view, the central open problem in moving from a front-end prior toward reliable end-to-end neural discovery[39–41]. Even short of that goal, DenoisedSR is immediately useful: it makes existing physical-law solvers both more successful and several times faster, with an empirically safe failure mode (recall $\geq$ 0.96 across every external suite we tested, and 1.000 on

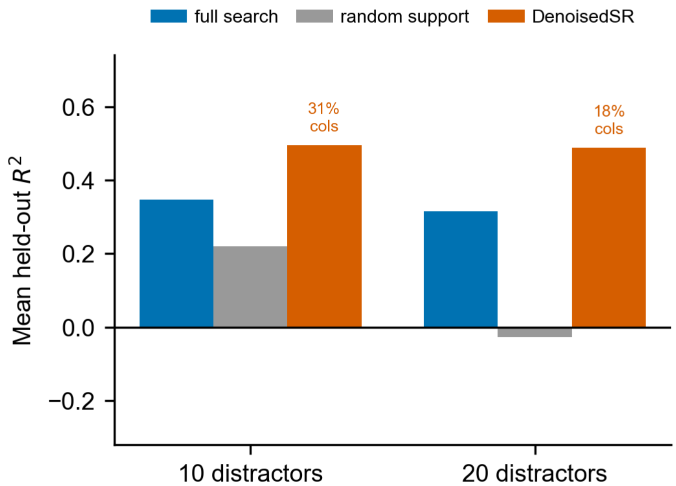

Fig. 8. The gain is search guidance, not fewer columns. Mean held-out R^2 on 24 PMLB/SRBench black-box tasks (three seeds, fixed 5 s budget) at 10 and 20 distractor variables. At matched, reduced column count (labels show columns relative to full search) DenoisedSR improves over full search while a same-size random support degrades far below it.

AI-Feynman across all 12 hard-scenario settings).

4 Methods

Problem setup. Each task provides $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{q}$ with $\mathbf{x}_i \in \mathbb{R}^m$, where only an unknown subset $S^\star \subseteq \{1, \ldots, m\}$ of columns appears in the target law $f^\star$. DenoisedSR predicts $\hat{S} \approx S^\star$ (and an optional operator set $\hat{O}$) and passes it to an SR backend run under a fixed time budget. Tasks are padded with random distractor columns to emulate the realistic regime in which a measurement table contains many irrelevant channels.

Front-end architecture. The variable-support predictor is an ensemble of two complementary models. (1) A graph-attention network[33,34] operates on a graph whose nodes are the input columns; node features combine per-column statistics (moments, correlation with the target, monotonicity and dimensional descriptors) and edges encode a variable co-occurrence prior learned from a knowledge graph of physical laws together with observation-derived correlation. Attention over this graph lets the model reason about which columns *jointly* support a plausible law rather than scoring columns in isolation. (2) A random forest scores each column from the same per-column features. The final score is $s(\text{col}) = \text{RF}(\text{col}) + \sigma(\text{GAT}(\text{col}))$, and a column is retained if $s \geq \tau$ with a low threshold ($\tau = 0.10$) chosen to favour recall. An analogous graph-attention model with graph-level pooling produces the multi-label operator prediction used for the optional operator prior.

Training. The predictors are trained only on internal physical-law records—formula templates with sampled observation tables drawn from a curated knowledge graph of physical laws—never on the external evaluation suites. Training pools of 8000–40 000 formulas were used, with observation tables pre-sampled across variable ranges and a 10–30 distractor curriculum so the front-end is robust to the number of irrelevant columns at test time. An adversarial variant (a generative-adversarial graph-attention model) was explored but gave no consistent gain and is not used.

Backends and evaluation protocol. The prior is verified against three backends from structurally different families. (a) PySR[5] (evolutionary genetic programming) is the primary backend because

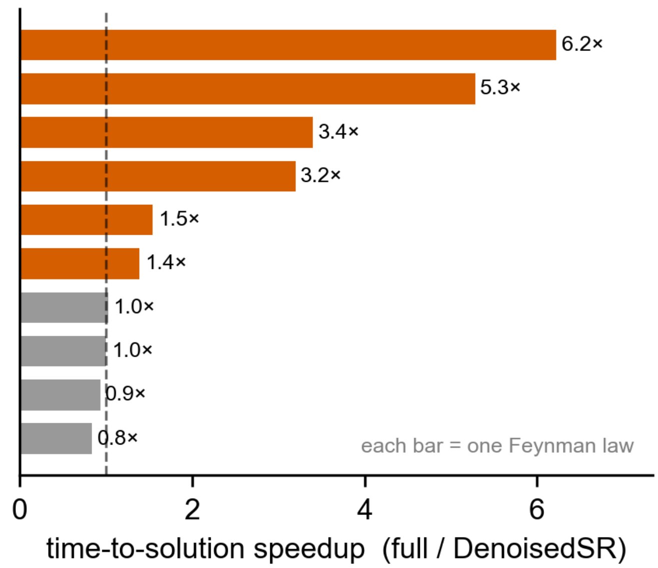

Fig. 9. Pruning accelerates time-to-solution. Per-formula speedup (full-search time divided by DenoisedSR time) with early stopping on exact recovery, on 10 Feynman formulas at a single seed. Non-trivial physics formulas solve up to 6× faster (median ∼1.5×); deep-tree formulas (depth-, not variable-limited) show little gain.

it exposes variable and operator restrictions directly and is reproducible locally. The headline 30-equation PySR comparison runs at a 10 s per-task budget, the full 17-operator library, with 20 random distractor columns at $q = 100$, repeated over three random seeds (a fourth seed crashed mid-run; mean±s.d. are reported over the three completed seeds). The distractor sweep $n_{\text{dist}} \in \{5, 10, 20, 30\}$ uses three seeds at $d \in \{5, 20, 30\}$ and two at d=10. The observation-count sweep $q \in \{50, 100, 200\}$ uses three seeds at $q \in \{50, 200\}$ and two at q=100, all at the deployed τ=0.10. (b) gplearn[25] (classical genetic programming, 20 generations, population 500, 10-operator library): three random seeds, identical other settings. (c) PSE[11] (parallel symbolic enumeration on GPU, the 2026 Nature Computational Science cover algorithm): three symbol layers, four $n_{\text{psrn\_inputs}}$ per round, nine-operator library, 30 s budget, single seed, run on an RTX 5080 GPU (the published default configuration of five inputs requires more VRAM than available). The same prior interface applies to enumeration and generative[22] backends not directly tested here.

Variable-support evaluation. Exact recovery denotes held-out $R^2 \geq 0.999$. Variable support quality is measured by recall against the known true-variable set at the deployed ensemble threshold $\tau = 0.10$, and by the *distractor filter rate*: the fraction of distractor columns the model correctly removes (zero would mean keeping all distractors; one would mean keeping none). Recall is the operationally critical quantity because dropping a true variable is fatal to downstream recovery. Five classical selector baselines (Pearson and Spearman $|\rho|$, mutual information, random-forest importance, and Lasso with cross-validated α) are scored on AI-Feynman at an oracle top-k budget that hands them the true number of variables. A controlled name-ablation that renames variables to $x_0, x_1, \ldots$ while keeping the data unchanged measures the marginal contribution of the physical-name prior in 12 hard scenarios spanning $q \in \{20, 50, 100\}$, noise $\eta \in [0, 1.0]$, distractor counts up to 100, and operating thresholds $\tau \in \{0.10, 0.50, 1.0\}$.

Operator-support evaluation. The operator predictor is a graph-attention model with graph-level pooling that outputs a multi-label probability over a 17-operator library; an operator is included if

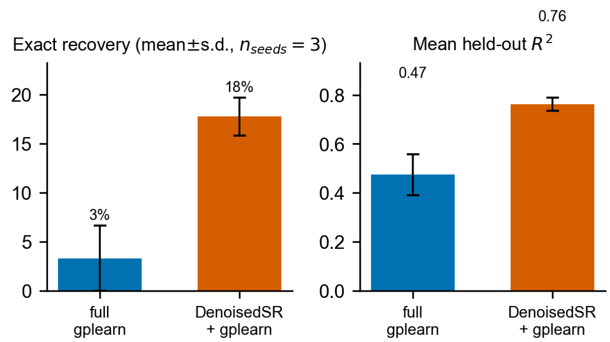

Fig. 10. Same gain transfers to a second backend. Exact recovery and mean held-out R^2 on the 30-equation Feynman subset with the gplearn backend[25] (20 generations, population 500, ~20 distractors; mean±s.d. over three random seeds). Full gplearn rarely recovers a law exactly (3.3±3.3%); supplied with the DenoisedSR variable prior over the same 3.4 predicted columns it recovers 17.8±1.9% exact, raising mean held-out R^2 from 0.48 to 0.76.

its probability exceeds $\tau_{\text{op}} = 0.30$ (deployed). Threshold sensitivity is reported over $\tau_{\text{op}} \in [0.03, 0.70]$ alongside the same name-ablation as for variables.

Robustness sweeps. Label-noise robustness is evaluated by adding $\eta \cdot \text{s.d.}(y_{\text{clean}})$ Gaussian noise to y for $\eta \in \{0, 0.01, 0.05, 0.10, 0.20, 0.30\}$ (up to 30% relative noise) and re-scoring all five baselines alongside DenoisedSR. The PMLB/SRBench black-box comparison uses 24 tasks at $q = 100$ with 5 s PySR budget, three random seeds, 10 and 20 distractor levels; the same-size random-support control retains the same number of columns selected without using the observations. Time-to-solution is measured with early stopping on exact recovery; front-end inference overhead is measured separately.

Reproducibility. All reported numbers are produced by the released evaluation scripts from the recorded result artifacts. The complete release (code, model weights, all backing JSON artifacts, paper TeX and figures) is archived on Zenodo at `10.5281/zenodo.20584889`.

Data availability

The external benchmarks (AI-Feynman, PMLB/SRBench, SRSD-Feynman) are public[2,8,24,42]; the SRSD-Feynman dummy variants used here are obtained from the authors' Hugging Face repositories. The physical-law training records are materialised from public formula and range information. The recorded result artifacts that back every figure and table are released with the code (Zenodo archive, `10.5281/zenodo.20584889`).

Code availability

Code for the front-end, training, evaluation and figure generation, together with the result artifacts and the trained graph-attention weights, is released at `https://github.com/ChenxiHeCam/DenoisedSR` under the MIT License, and archived on Zenodo at `10.5281/zenodo.20584889`. The two large random-forest partner weights (~213 MB and ~341 MB), which exceed GitHub's per-file limit, are bundled with the Zenodo archive.

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Author contributions

C.H. conceived the study, designed and implemented the front-end and the evaluation pipeline, performed the experiments and analysis, and wrote the manuscript. J.C. supervised the project, advised on the methodology and the physics evaluation, and revised the manuscript. Both authors approved the final version.

Competing interests

The authors declare no competing interests.

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Supplementary information

Table 2. Summary of variable-support recovery across selectors and suites. Variable-support recall, with the fraction of tasks on which the selector attains *perfect* recall (every true variable retained). All selectors except DenoisedSR are run at an oracle top-k budget that hands them the true variable count for free. Recall on AI-Feynman is also reported under relative Gaussian label noise η; entries marked — were not part of the noise sweep.

	AI-Feynman (118)		SRSD (120)	AI-Feynman recall under noise η				
Selector	recall	perfect %	recall	0.01	0.05	0.10	0.20	0.30
Pearson $\lvert\rho\rvert$	0.768	45.8	—	0.772	0.769	0.771	0.763	0.768
Spearman $\lvert\rho\rvert$	0.776	45.8	—	0.773	0.770	0.770	0.757	0.754
mutual information	0.634	20.3	—	0.633	0.636	0.644	0.608	0.605
random-forest importance	0.812	44.9	—	0.815	0.822	0.815	0.807	0.799
Lasso (CV) *oracle k*	0.905	71.2	0.723	0.904	0.904	0.886	0.881	0.868
DenoisedSR (ours, τ=0.10)	**1.000**	**100.0**	**0.967**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**

Table 3. Threshold sensitivity of the deployed RF+GAT ensemble. Sweep of τ on AI-Feynman (118 laws, $q = 100$, 20 distractors). Recall remains 1.000 with 100% perfect-recall over $\tau \in [0.05, 0.30]$ (a sixfold range); only at $\tau \geq 0.50$ does recall start to decline. The deployed $\tau = 0.10$ is not cherry-picked.

τ	Recall	Precision	Perfect-recall (%)	Mean kept columns
0.05	1.000	0.705	100.0	9.03
0.08	1.000	0.740	100.0	8.81
0.10 (deployed)	**1.000**	**0.749**	**100.0**	**8.76**
0.15	1.000	0.766	100.0	8.67
0.20	1.000	0.769	100.0	8.66
0.30	1.000	0.780	100.0	8.60
0.50	0.998	0.873	99.2	6.31
0.75	0.981	0.991	93.2	3.87
1.00	0.959	1.000	86.4	3.72

Table 4. Internal validation under a different protocol (not directly comparable to the four external benchmarks of Fig. 2). Evaluated on internally-curated physical-law records drawn from the same knowledge graph as the training pool, using each record's pre-sampled observation values rather than freshly sampled tables, and without random distractor padding. We report it as a separate scale check rather than a fifth external suite.

Set	Equations	Recall	Perfect-recall (%)
Internal physics-law set (v5_physics65)	1085	0.999	99.7
Internal OOD-91 curated subset	91	0.996	98.9

Table 5. Variable filter is insensitive to physical names. Same data, same model, same threshold $\tau = 0.10$; only the variable names are changed between "named" (physical: m, v, q, ϵ, ...) and "anonymized" (x_0, x_1, ...). Across 12 scenarios on AI-Feynman 118, the physical-name prior contributes at most 0.3 percentage points of variable filter rate. The deployed RF+GAT ensemble is dominated by the random-forest column-statistics head; the graph-attention head's co-occurrence-graph contribution is largely vestigial for the variable task.

	Named (physical)		Anonymized (x_i)		Δ (named−anon)	
Scenario	recall	filter%	recall	filter%	recall	filter (pp)
baseline q=100, d=20	1.000	98.4	0.998	98.3	+0.002	+0.04
small q=50	1.000	97.0	0.998	97.0	+0.002	+0.04
very small q=20	1.000	90.8	0.998	90.6	+0.002	+0.18
noise $\eta = 0.30$	1.000	98.7	0.998	98.6	+0.002	+0.08
noise $\eta = 0.50$	1.000	98.6	0.998	98.5	+0.002	+0.10
noise $\eta = 1.00$	1.000	98.6	0.998	98.6	+0.002	+0.07
many distractors d=50	1.000	98.5	0.998	98.5	+0.002	+0.04
many distractors d=100	1.000	98.6	0.998	98.5	+0.002	+0.09
hard q=20, $\eta = 0.50$	1.000	90.9	0.998	90.7	+0.002	+0.27

Table 6. Operator filter *is* sensitive to physical names. Same data, same operator predictor, varying threshold τ_{op}. Operator filter rate is the fraction of the 13 non-trivial operators (excluding the four always-on arithmetic operators) dropped from the search library. Physical names contribute +7 to +17 percentage points of operator recall—a much larger effect than for variables, because the operator-vs-data signal in $(\mathbf{x}, y)$ is weak and the physics-knowledge-graph prior is the main source of information.

τ_{op}	Named (physical)			Anonymized (x_i)			Δ recall
	recall	prec.	filter%	recall	prec.	filter%	(named−anon)
0.03	0.912	0.383	58.9	0.744	0.346	59.8	+0.168
0.10	0.825	0.465	67.9	0.726	0.380	65.6	+0.099
0.20	0.761	0.503	72.9	0.703	0.427	70.1	+0.058
0.30 (deployed)	**0.738**	**0.536**	**76.5**	**0.667**	**0.437**	**72.9**	+0.071
0.50	0.597	0.547	82.7	0.521	0.420	82.2	+0.076
0.70	0.407	0.532	88.9	0.285	0.376	90.4	+0.122

Table 7. Backend integration summary. Mean exact-law recovery and mean held-out R^2 on the 30-equation Feynman headline subset ($n_{\text{dist}} = 20$, $q = 100$). Errors are ±s.d. over three random seeds. The variable prior shrinks the candidate-column set from 23.2 to 3.4 in both backends.

Condition	Exact recovery (%)	Mean held-out R^2
PySR (10 s budget, full 17-op library)		
full search	42.2 ± 13.5	0.906 ± 0.045
+ DenoisedSR variable prior	60.0 ± 0.0	0.980 ± 0.011
+ DenoisedSR variable + operator prior	60.0 ± 11.5	0.979 ± 0.011
gplearn (20 generations, population 500, 10-op library)		
full search	3.3 ± 3.3	0.475 ± 0.083
+ DenoisedSR variable prior	17.8 ± 1.9	0.763 ± 0.029
PSE (parallel GPU enumeration, 30 s budget, 9-op library, single seed)		
full search	0	0.184
+ DenoisedSR variable prior	40	0.895

Table 8. Master backend × benchmark matrix. Exact-law recovery (% with held-out $R^2 \geq 0.999$) on five physical-symbolic benchmarks for three structurally different SR backends, with and without the DenoisedSR variable prior. Cells show *full backend* → *+DenoisedSR*; gap in percentage points. Multi-seed entries are mean±s.d.; single-seed entries omit the standard deviation. PySR-Feynman118 is over three random seeds; gplearn entries on Feynman are over three seeds; remaining cells are single seed. PSE-SRSD is not evaluated because PSE's float32 tensor representation underflows on SRSD's $\sim 10^{60}$-magnitude physical-unit targets (a backend-data interaction, unrelated to the prior). Bold marks the larger of the two values in each row.

Benchmark	Distractors	PySR *full* → *+prior* (Δ)	gplearn *full* → *+prior* (Δ)	PSE *full* → *+prior* (Δ)
AI-Feynman 118 (3 seeds)	20 random	46.3±3.0 → **57.3±1.8** (+11)	0.8 → **11.0** (+10)	8.5 → **42.4** (+34)
Strogatz 15	20 random	60.0 → **66.7** (+7)	0.0 → 0.0 (0)	6.7 → **53.3** (+47)
Nguyen 12	20 random	**91.7** → **91.7** (0)	8.3 → 8.3 (0)	33.3 → **100.0** (+67)
SRSD-Feynman 120	~2 plausible	**54.2** → 50.0 (−4)	30.0 → 30.0 (0)	—
SRSD-Feynman 120	+ 20 random	42.5 → **54.2** (+12)	25.0 → **29.2** (+4)	—

Table 9. Headline result scales to 80 Feynman equations. Repeating the PySR comparison at the same configuration ($n_{\text{dist}} = 20$, $q = 100$, $\tau = 0.10$, single seed) on the first 80 Feynman tasks (chosen by $2 \leq |\text{features}| \leq 9$) rather than the headline 30. The gain is preserved at larger scale; full PySR's higher absolute rate at this scale (54% vs. 30–57% across seeds at n=30) is consistent with the 30-equation subset including a higher fraction of harder formulas.

Condition	Exact (%)	Mean held-out R^2	Variable recall
full PySR (all 23 cols, 17 ops)	54	0.900	—
+ DenoisedSR variable prior (3.4 cols)	66	0.981	1.000
+ DenoisedSR variable + operator prior	68	0.981	1.000

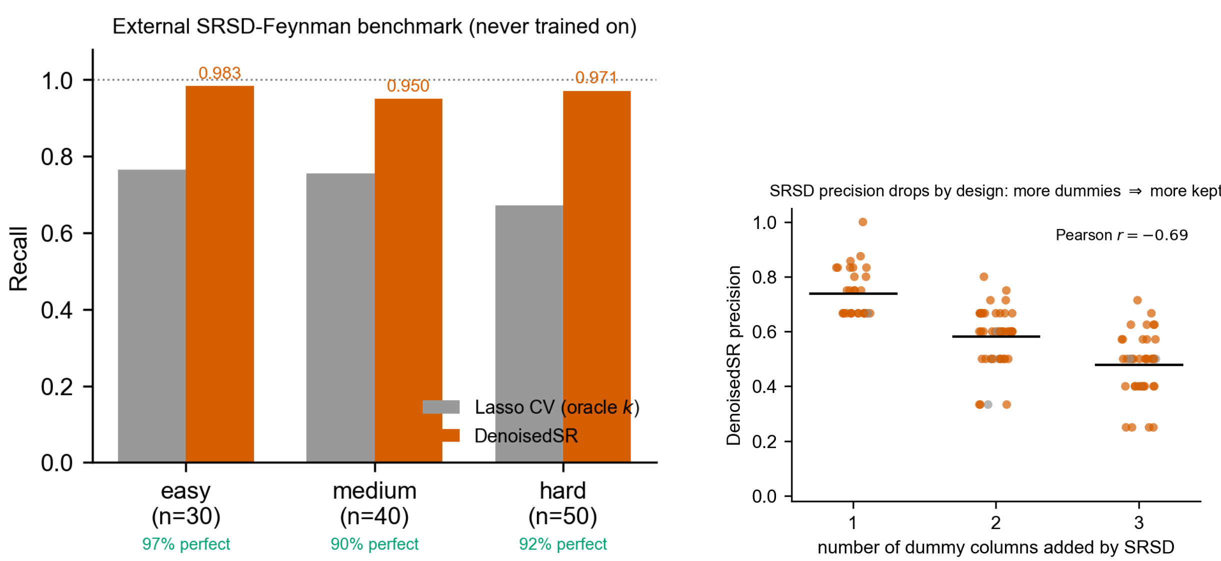

Fig. 11. Detail on SRSD-Feynman. *Left*: recall on each SRSD difficulty split (30 easy, 40 medium, 50 hard) for DenoisedSR vs. Lasso at oracle top-k; DenoisedSR exceeds 0.95 recall on every split. *Right*: diagnostic of the apparent precision drop on SRSD (0.59 overall). Each marker is one of the 120 SRSD tasks (orange = perfect recall, grey = imperfect recall); horizontal bars show the binned mean precision. Precision falls monotonically with the number of dummy columns SRSD adds to the task (Pearson $r = -0.69$) because the front-end at τ=0.10 retains physically plausible dummies rather than discarding them; on tasks with no dummies our precision is near 1.